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Detection of members of the *Klebsiella pneumoniae* complex by real-time PCR
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Goal: Real-time PCR detection of bacteria belonging to *K. pneumoniae* (Kp) and related species (*Kp1* to *Kp7* phylogroups, making up the Kp complex) by specific amplification of an intergenic sequence of 82 bp between *zur* (zinc uptake regulation protein) and *khe* (annotated as coding haemolysin).

Name of PCR: zikr (zinc-khe intergenic region)

Primers: zkir_F and zkir_R

zkir_F: 5' CTAAAACCGCCATGTCCGATTAA 3'

zkir_R: 5' TTCCGAAAATGAGACACTTCAGA 3'

Thermocycler: StepOnePlus™ Real-Time PCR System (Thermo Fischer Scientific)

PCR mix:

• **Master Mix:** Takyon™ ROX SYBR® MasterMix 2X Blue dTTP 7.5 ml (Ref. UF-RSMTB0705, Eurogentec)

According to the thermocycler used, Takyon™ Low ROX SYBR® MasterMix 2X Blue dTTP may be used

• **T4 bacteriophage Gene 32 product:** T4 gp32 (500 µg/ml) (Ref. 11TGP32100, MP Biomedicals) Prevents PCR inhibition for DNA extracted from complex matrixes such as soil; improves *Taq* DNA Polymerase amplification efficiency

Mix reagents	Volume per well (µl)
Takyon™ ROX SYBR® MasterMix 2X	10
Forward (3 µM)	2
Reverse (3 µM)	2
T4 gp32*	0.5
DNA template	2.5
PCR grade water	QS 20 µl

*optional

Amplification programme:

	Temperature (°C)	Time (minutes:seconds)	
Holding stage	95.0	03:00	Enzyme activation
Cycling stage (repeat 40x)	95.0	00:10	Annealing (data collection)
	60.0	01:00	
Melt Curve stage	95.0	00:15	Dissociation stage Program Step and Hold T° increment +0.3°C
	60.0	01:00	
	95.0	00:15	

DNA template: For environmental matrixes such as soil, 500 µl of 24h-enrichments (e.g., in LB with 10 mg/L ampicillin) are centrifuged (5 min., 5 800 x g) and washed with sterile water before boiling for 10 min. Boiled suspensions are then diluted 1/10 and 1/100 for qPCR.

Positive and negative controls:

Negative control: PCR grade water

Positive control: *Kp1* DNA (*Klebsiella pneumoniae* ATCC 700603) at concentration 2 ng/µl ; 2.5 µl used *i.e.*, 5 ng.